

MK11 to get Closed Beta Access

Mortal Kombat 11 fans will have a chance to play Mortal Kombat 11 a month before its official release <https://digit.in/apr19-47>



Baby's First Word Is 'Google'

Los Angeles-based parents used Google's Home smart speaker to play songs for their child, their baby's first word was Google <https://digit.in/apr19-48>

Quantum entanglement

It is the two qubits which are in a superposition which are correlated with each other, which means that the state of one qubit depends on another qubit. For example, consider the qubits as identical twins which have the same properties. So when one twin gets hurt the other one gets hurt too.

HOW DO THESE PROPERTIES ALTER THE PERFORMANCE OF THE QUANTUM COMPUTER?

Superposition and entanglement can enable quantum-computer to function very fast by working like a highly complex switch. With the help of Superposition and entanglement, quantum computers can work on highly advanced math problems.

To exemplify how these properties makes quantum computers more powerful let's talk in terms of number. A normal computer works in the produced system states in terms of 2^n where n represents the bits. So if $n = 10$, the number of bits produced will be 20. So a normal computer can represent only one system state at a time.

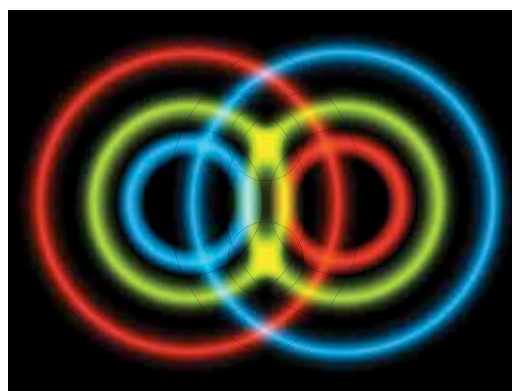
Now coming to quantum computers, it will have the power to represent qu-bits in 2^n system states. This can enable quantum computers to perform parallel operations on all the system states at once. So if $n = 10$, this means that whenever a qubit is added to the quantum computer it keeps becoming able to perform multiple tasks at once. In this case it could work on 1024 system states at once whereas a normal computer could only work on only one system state at a time.

For example, imagine a maze. So, if a normal computer was solving the puzzle, it would check all the possible ways to complete the puzzle by a trial and error method. But if a quantum computer was solving the puzzle, it would use its entangled quantum states to solve the maze way quicker than the normal computer.

This is what enables quantum computers to solve problems way efficiently than normal computers.

HOW LONG WILL IT TAKE FOR QUANTUM COMPUTERS TO COME INTO EXISTENCE?

The race towards building a quantum computer has already begun. Tech giants such as Intel, Google and IBM are working on packing more qubits into their quantum computers to establish a better performance. Quantum computers will not be commercially viable for any time soon due to two main



Two qubits in a superposition are said to be quantum entangled

reasons. The first reason is computational power. Among the quantum computers which have been built lately, Google has the most powerful one with 72 qubits.

Google is very close to a milestone in which these machines will be able to build a computer which can do certain things that normal computers cannot. This milestone is known as "quantum supremacy". As a rival, Multinational tech biggie D-Wave is working on a 4000 qubit quantum computer which costs 15 million dollars.

The second reason is systematic errors. Qubits are very fragile as it deals with subatomic particles and the quantum realm itself. The Qubits needs to be kept in very cold environment to keep them functioning. So scientists could only run algorithms for seconds until the qubits fall out of the quantum state. These errors need to be solved by adding more qubits in their place. By doing these things quantum computers consume a lot of computational power which literally translates to electricity.

WHAT WILL QUANTUM COMPUTERS BE USED FOR?

Quantum computers can solve problems much faster than conventional computers, which makes them particularly suitable for industries that depend on making timely predictions. This is why quantum computers can be used in financial services to model financial data and prevent risk factors to make better investments.

Another industry where quantum computers will be used is medicines and materials. Researchers in Harvard university puzzled one of the quantum computers to find out the complex folds of proteins. The model of proteins consisted of chains of amino acids which involved mathematics. This puzzle was tried with a 49 qubit quantum computer and a 69 qubit quantum computer but the answers were not accurate.

But when it was tried with a 71 qubit quantum computer the answer given was 13 which were accurate. It was able to accomplish an amazing feat by modelling the behaviour of protein folding with some degree of accuracy.

From managing money to massive datasets, the possibilities of quantum computers are endless. Advancement towards machine learning could drastically increase the efficiency of how the world functions. Discoveries of new medicines and materials could lead to a peaceful and secure world. The enhanced financial services can drastically change the way we invest.

With scientists, tech companies and physicists working on quantum computers to perfect this technology, these truly mind blowing uses of quantum computers could become universal. Until then let's stay fastened in our seats to converge into the quantum age.

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